# How Much Personal History Is Enough?

*Measuring the data requirement for individualised financial behavioural anomaly detection*

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**QUESTION** **How many of a person's own transactions** does a detector need before the personal baseline is worth using?

| 25 | 10 | 96% → 1.2% | 5 of 5 |
|---|---|---|---|
| transactions reach 90% of best PR-AUC (Berka) | transactions: the personal model starts to beat the crowd | flagged at n = 10: in-sample vs leave-one-out threshold | independent cohort draws give the same 25-transaction threshold |

## 1 Motivation

A personalised detector scores each transaction against **one person's own history**. When an account opens, that history is empty.

- Deployed systems filter out "unseasoned users"
- The cutoff is assumed, not measured
- We found no published measurement of it
- Thin histories: new customers, young people, recent immigrants

**Q:** Where does the personal baseline become worth using?

## 2 Protocol

Hold a 60-transaction window fixed. Train on the **n** transactions just before it.

| TRAIN | HOLD OUT | LABEL | READ OFF |
|---|---|---|---|
| n personal transactions | the same 60-transaction window | swap in a stranger's transaction | smallest n reaching 90% |

- Paired: every n scored on the same window
- Labels by provenance, not by any detector
- 10% injected: six anomalies per window
- Population model: fitted on 400 donor accounts

## 3 Three data layers

| Layer | Entities | Nature |
|---|---|---|
| **Berka bank** | **300 accounts** | **real** |
| Simulation | 240 people | controlled |
| TabFormer | 300 entities | synthetic |

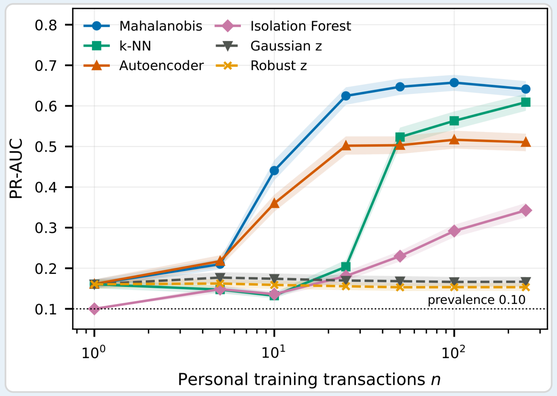
Six unsupervised detectors on each layer: Mahalanobis, k-NN, isolation forest, autoencoder and two z-scores.

## 4 Limits and ethics

- One bank, 1993–1998; the second dataset is synthetic
- A flag is a deviation, not fraud
- District-level audit: no Holm-significant false-positive gap; age, tenure and immigration untested

## 5 Sufficiency point ★ Headline

Berka, 300 accounts, impostor labels. Prevalence baseline 0.10.




**PR-AUC against n.** Mahalanobis rises steeply and saturates; k-NN jumps between 25 and 50; both z-scores never learn.

| 25 | 0.657 | 98% |
|---|---|---|
| transactions to reach 90% of best | best PR-AUC (Mahalanobis) | of the gain arrives by n = 50 |

## 6 Three definitions, three answers

| Detector | $y_{max}$ | τ | Rel-90 | Rel-95 |
|---|---|---|---|---|
| **Mahalanobis** | **0.658** | **11.5** | **25** | **25** |
| Autoencoder | 0.518 | 11.1 | 25 | 25 |
| k-NN | 0.633 | 50.9 | 100 | 250 |
| Isolation forest | 0.350 | 71.2 | 250 | 250 |
| Robust z | 0.153 | 21.7 | 1 | — |
| Gaussian z | 0.170 | 0.25 | 1 | — |

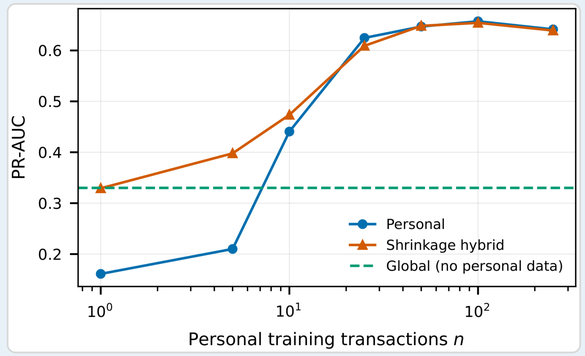
- Rel-90: smallest n reaching 90% of best
- Stability criterion is never met by a detector that learns
- Rel-90 rewards flat curves: robust z gives n = 1

## 7 Recency beats volume

| 0.647 → 0.642 | −0.036 | ρ = −0.08 |
|---|---|---|
| 50 to 250 transactions, natural density | same 50 drawn from a 4× longer span | drift vs. loss (Holm p = 0.32) |

## 8 Crowd or person? ★ KEY

Below a crossover, a model built from **400 other accounts** beats the person's own history.




**Mahalanobis, Berka.** Personal falls below global until n = 10; the hybrid tracks or beats both.

**SHRINKAGE HYBRID (K = 25, FIXED IN ADVANCE)**

$$\mu_h = \frac{n\,\mu_p + \kappa\,\mu_g}{n + \kappa}$$

| n | weight on the person | κ | weight on the crowd |
|---|---|---|---|
| n = 0 | the global model | n large | the personal model |

Crossover at n = 10 (Mahalanobis) and n = 50 (k-NN).

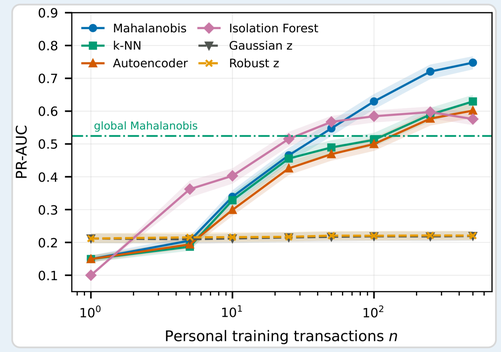
## 9 Calibration moves the operating point

Mahalanobis, nominal false-positive rate (FPR) 0.05. LOO = leave-one-out.

| n | In-sample flag rate | LOO flag rate | LOO FPR | LOO recall |
|---|---|---|---|---|
| 5 | 0.989 | 0.152 | 0.151 | 0.163 |
| 10 | **0.960** | **0.012** | 0.007 | 0.058 |
| 25 | 0.391 | 0.110 | **0.066** | 0.503 |
| 50 | 0.251 | 0.126 | 0.073 | 0.612 |
| 100 | 0.196 | 0.134 | 0.076 | 0.658 |
| 250 | 0.162 | 0.136 | 0.080 | 0.647 |

- In-sample scores are biased low by construction
- LOO is silent at n = 10, near-nominal from n = 25

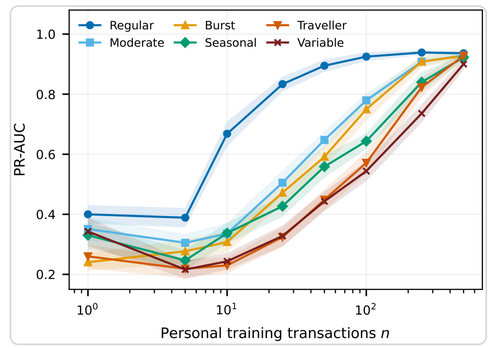
## 10 A second dataset reverses a ranking




**TabFormer (synthetic).** Same shape, later threshold; isolation forest is best at n = 5 to 50.

- Mahalanobis threshold is 250, not 25
- Isolation forest: worst on Berka, best at small n here
- Crossover moves to n = 25–100 (stronger global model)

## 11 Variability sets the rate




**Simulation.** Regular people saturate by n = 50; travellers and variable people need about 500.

- Under controlled manipulation: ρ = 0.48–0.71, all Holm p < 0.001
- In real Berka accounts no descriptor predicts it (|ρ| ≤ 0.12)

## 12 Takeaways

**IDEA.** A personal baseline ranks well from about 25 transactions.

**METHOD.** Provenance labels, paired windows, three data layers.

**RESULT.** A hybrid removes the cold-start penalty; calibration moves the operating point.

**PRACTICAL.** Shrink toward the crowd, keep recent history, calibrate out-of-sample.



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